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System for fiber connector assembly

Abstract

An example system for fiber connector assembly is provided. The system includes an adapter configured to at least partially receive a plug housing subassembly. The system includes a base station configured to receive the adapter, the plug housing subassembly, and a first backshell component. The base station includes structural features designed to align and maintain a position of the plug housing subassembly relative to the first backshell component prior to securing a second backshell component to the first backshell component.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of a commonly assigned U.S. Provisional Patent Application No. 63/331,995, which was filed on Apr.

18, 2022. The entire content of the foregoing provisional application is incorporated herein by reference.

BACKGROUND

(1) A variety of connectors are used with fiber optic cables in the telecommunications industry. During manufacturing, the fibers of the fiber optic cable are generally passed into backshell components with an operator positioning each of the ferrules in the desired positions. Manually positioning the ferrules in the proper position in an accurate and consistent manner can be difficult, resulting in potential inconsistencies among the assembled connectors. Such inconsistencies can result in undetectable damage that leads to poor performance and/or reliability issues.

SUMMARY

(2) Embodiments of the present disclosure provide an exemplary system for fiber connector assembly. The system includes a stub adapter capable of receiving a plug housing and an internal brace, a base station that precisely positions/aligns the lower backshell half and the stub adapter with the terminated components, and a hand tool for delicate and accurate maneuvering of the brace to seat it properly in the lower backshell half. The system allows for repeatedly accurate assembly of the fiber connector, reducing the risk of component damage or erroneous assembly. In some embodiments, the base station can be placed on a rotatable index plate to allow operators to turn the base station to an orientation most comfortable to the operator, thereby further improving the accuracy of the assembly process. Although discussed herein as used to assemble an 8-fiber alignment independent multifiber (AIM) trunk connector, it should be understood that the system could be adapted for assembly of other types of fiber connectors.

(3) In accordance with embodiments of the present disclosure, an exemplary system for fiber connector assembly is provided. The system includes an adapter configured to at least partially receive a plug housing subassembly. The system includes a base station configured to receive the adapter, the plug housing subassembly, and a first backshell component. The base station includes structural features designed to align and maintain a position of the plug housing subassembly relative to the first backshell component prior to securing a second backshell component to the first backshell component.

(4) The plug housing subassembly can be a terminated plug housing subassembly including a plug housing and an internal brace. The adapter can include a rear housing detachably coupled and/or engaged with a front housing. The front housing is configured to at least partially receive the plug housing subassembly. The front housing can include a body with an opening formed in a front face of the body and extending into a hollow interior of the body. The front housing can include an array of alignment sleeves (e.g., 8 sleeves, or the like) extending from a rear wall within the hollow interior of the body. Each of the alignment sleeves of the array of alignment sleeves configured to receive and support a fiber ferrule. In some embodiments, each alignment sleeve of the array of alignment sleeves can include an inner surface fabricated from, covered by or coated with zirconia ceramic or phosphor bronze.

(5) The base station can include a primary section and a secondary section extending from the primary section. The primary section can define an enclosure configured to at least partially receive the adapter. The primary section can include a base with raised blocks extending vertically on opposing sides of the base. The raised blocks can define a narrowed section within the enclosure complementary to outer surfaces of the adapter. The base station can include a raised step along a base surface at a transition between the primary section and the secondary section. The raised step is configured to abut a front face of the adapter and an inner surface of a rear wall of the primary section is configured to abut a rear face of the adapter during assembly of the adapter with the base station.

(6) The secondary section can include a backshell support including a cable support section and outwardly flanged walls extending from the cable support section. The outwardly flanged walls

define interior surfaces complementary to outer surfaces of the first backshell component. The outwardly flanged walls can face the primary section of the base station. The secondary section can include raised platforms on opposing sides of a top surface. The raised platforms are spaced by a gap. The gap is dimensioned complementary to a section of the first backshell component to ensure alignment of the first backshell component relative to the base station during assembly.

(7) The base station can include a cover pivotably coupled to the primary section. The cover is configured to abut a front face of the adapter to maintain alignment and positioning of the adapter and the plug housing subassembly relative to the first backshell component. Pivoting of the cover into a closed position over the adapter simultaneously prevents movement of the adapter, the plug housing subassembly, and the first backshell component relative to the base station. In some embodiments, the system can include a hand tool for manipulating a position of one or more components (e.g., an internal brace) of the plug housing subassembly at least partially received by the adapter. In such embodiments, the tool can include a handle and a distal end with a V-shaped cutout complementary to an edge of the one or more components of the plug housing subassembly

(8) In accordance with embodiments of the present disclosure, an exemplary system for fiber connector assembly is provided. The system includes an adapter configured to at least partially receive a plug housing subassembly. The system includes a first backshell component configured to at least partially receive the plug housing subassembly. The system includes a second backshell component configured to at least partially receive the plug housing subassembly. The system includes a base station configured to receive the adapter, the plug housing subassembly, and the first backshell component. The base station includes structural features designed to align and maintain a position of the adapter and the plug housing subassembly relative to the first backshell component prior to securing the second backshell component to the first backshell component.

(9) In accordance with embodiments of the present disclosure, an exemplary method of fiber connector assembly is provided. The method includes at least partially positioning a plug housing subassembly into an adapter of a fiber connector assembly system. The method includes positioning a first backshell component of a fiber connector onto the base station of the fiber connector assembly system. The method includes positioning the adapter and the plug housing subassembly onto the base station. The method includes aligning a position of the plug housing subassembly relative to the first backshell component. The method includes maintaining the aligned position of the plug housing subassembly relative to the first backshell component with structural features of the base station prior to securing a second backshell component to the first backshell component

(10) Any combination and/or permutation of embodiments is envisioned. Other objects and features will become apparent from the following detailed description considered in conjunction with the accompanying drawings. It is to be understood, however, that the drawings are designed as an illustration only and not as a definition of the limits of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To assist those of skill in the art in making and using the system for fiber connect assembly, reference is made to the accompanying figures, wherein:

(2) FIG. 1 is a perspective view of an adapter in an exemplary embodiment;

(3) FIG. 2 is a front view of an adapter of FIG. 1;

(4) FIG. 3 is a perspective view of an adapter of FIG. 1 including a terminated plug subassembly;

(5) FIG. 4 is a perspective view of a base station in an exemplary embodiment;

(6) FIG. 5 is a top view of a base station of FIG. 4;

(7) FIG. 6 is a side view of a base station of FIG. 4;

- (8) FIG. 7 is a rear view of a base station of FIG. 4;
 - (9) FIG. 8 is a perspective view of a base station of FIG. 4 including a backshell half;
 - (10) FIG. 9 is a perspective view of a base station of FIG. 4 in an open position, the base station including a backshell half and an adapter/terminated plug subassembly of FIG. 3;
 - (11) FIG. 10 is a perspective view of a base station of FIG. 4 in a closed position, the base station including a backshell half and an adapter/terminated plug subassembly of FIG. 3;
 - (12) FIG. 11 is a perspective view of a positioning tool in an exemplary embodiment;
 - (13) FIG. 12 is a detailed view of a positioning tool of FIG. 11;
 - (14) FIG. 13 is a perspective view of an assembly of a base station assembly of FIG. 10 and a positioning tool of FIG. 11; and
 - (15) FIG. 14 is a perspective view of an assembled fiber connector in an exemplary embodiment.
- DETAILED DESCRIPTION

(16) The system for fiber connector assembly discussed herein provides an assembly tool station that combines multiple tools and fixtures together in a single system that provides for consistent assembly of fiber connectors, such as the AIM connector. The system provides for an efficient and accurate assembly of such fiber connectors, resulting in a simplification and improvement of the connector assembly process (as compared to traditional assembly by operators). The system generally includes an adapter (e.g., a stub adapter) configured to receive a terminated plug housing and an internal brace. In some embodiments, the adapter can receive a terminated subassembly including the plug housing and the internal brace. The system includes a base station configured to receive the assembly of the adapter and the terminated subassembly. The base station is further configured to receive a backshell half of the connector. The base station includes structural features and/or components that ensure alignment of the adapter, the terminated subassembly, and the backshell half relative to each other, and includes a hinged top or cover that maintains the position/alignment of the assembled components relative to each other.

(17) The system includes a positioning hand tool with a tapered end to catch on the rear edge of the internal brace of the terminated subassembly, enabling delicate maneuvering of the internal brace to seat the brace properly in the backshell half supported by the base station. Using the noted components or elements of the system, the repeated accurate assembly of the fiber connector is simplified and ensured, reducing the risk of component damage and/or erroneous assembly. In some embodiments, the base station can be placed on a rotatable index plate to allow the individual operators to turn the base station to an orientation that is most comfortable to the operator, providing greater flexibility and accuracy in the assembly/manufacturing process.

(18) FIGS. 1-2 are perspective and front views of an exemplary adapter **100** of a system for fiber connector assembly of the present disclosure. When used for assembly of an AIM fiber connector, the adapter **100** is configured as an AIM-to-4xMDC conversion adapter. However, when used for assembly of different fiber connectors, the adapter **100** can include an interface complementary to such other fiber connectors. The adapter **100** includes a rear housing **102** detachably coupled to a front housing **104**. The rear housing **102** generally includes a body with a top surface **106**, an opposing bottom surface **108**, opposing side surfaces **110**, **112**, and opposing rear and front faces **114**, **116**. The rear housing **102** includes two flanges **118**, **120** extending from the side surfaces **110**, **112** and aligned with the front face **116**, such that the front face **116** width is dimensioned greater than the width of the rear face **114**.

(19) In some embodiments, the rear housing **102** can define a substantially T-shaped configuration. It should be understood that the rear housing **102** can define a different configuration and does not need to define a specific form, so long as the rear housing **102** can engage with the front housing **104**. In addition, the rear housing **102** does not need to mimic the 4xMDC interface, and was instead used in the present embodiment due to a large number of convenient indexing surfaces provided by the rear housing **102**. The configuration of the rear housing **102** can be selected to include structural elements and/or contours that assist with accurately locating the adapter **100**

within the base station.

(20) The front face **116** of the rear housing **102** includes an opening (not visible due to assembly with front housing **104**) formed therein, the opening complementary to a rear extension of the front housing **104**. The front housing **104** generally includes a body with a top surface **122**, an opposing bottom surface **124**, opposing side surfaces **126**, **128**, and opposing rear and front faces **130**, **132**. The front housing **104** generally defines a rectangular block configuration. In some embodiments, the side surfaces **126**, **128** can each include a latch **134**, **136**. In some embodiments, the latches **134**, **136** can assist in indexing the adapter **100** relative to the base station during the assembly process. In some embodiments, the front housing **104** can be free of latches **134**, **136**. The front housing **104** can include an extension (not visible) protruding from the rear face **130** and complementary to the opening in the front face **116** of the rear housing **102** to assist with alignment of the front housing **104** relative to the rear housing **102** during assembly. The front housing **104** can be slidably removed from the rear housing **102** as needed, and can generally be maintained in the assembled configuration via, e.g., friction.

(21) The front housing **104** includes an opening **138** formed in the front face **132** and extending into a hollow interior **140**. The opening **138** generally defines a rectangular perimeter, with one side of the opening **138** including an outwardly extending alignment groove **142**. The groove **142** corresponds with a complementary protrusion in the connector plug housing to be inserted into the opening **138**, ensuring the correct orientation of the connector plug housing during assembly. The hollow interior **140** extends to a rear inner wall **144**. The front housing **104** includes an array of alignment sleeves **146** extending from the rear inner wall **144** towards the front face **132**. In the embodiment of FIGS. 1-2, the front housing **104** includes eight alignment sleeves **146** to correspond with eight ferrules of a corresponding AIM connector. However, the front housing **104** can include a greater or lower number of sleeves **146** depending on the type of connector being assembled.

(22) Each alignment sleeve **146** includes an opening **148** extending therethrough to receive and hold the terminated ferrules in the plug housing to be assembled with the adapter **100**. The openings **148** ensure that the ferrules are maintained straight and secure during the assembly process. The openings **148** and/or alignment sleeves **146** can be fabricated from zirconia ceramic or phosphor bronze. After extended usage of the adapter **100**, the zirconia ceramic or phosphor bronze may wear over time. In some embodiments, if such wear occurs and/or if the alignment sleeves **146** are damaged, the front housing **104** can be easily disengaged from the rear housing **102** and replaced with a new front housing **104** that can be engaged with the rear housing **102** for continued use. In some embodiments, if such wear occurs and/or if the alignment sleeves **146** are damaged, the entire adapter **100** can be replaced (e.g., the rear and front housings **102**, **104** cannot be disengaged from each other).

(23) FIG. 3 is a perspective view of the adapter **100** and a terminated plug subassembly **150** inserted into the adapter **100**. The subassembly **150** generally includes a plug housing **152** and an internal brace **154** inserted and/or coupled relative to the plug housing **152**. The plug housing **152** can including opposing side walls **156**, **158**, a bottom wall **160** connecting the side walls **156**, **158**, and a top surface defined by first and second top wall sections **162**, **164** connecting the side walls **156**, **158**. The first and second top wall sections **162**, **164** are spaced from each other by a gap or notch formed in the top surface. However, the top wall sections **162**, **164** extend along the same plane and are structurally connected by a central wall section (not visible) such that the top surface forms a single structure. The top wall section **162** extends substantially perpendicularly from the side wall **156** towards the top wall section **164**, and the top wall section **164** extends substantially perpendicularly from the side wall **158** towards the top wall section **162**.

(24) The plug housing **152** includes a tab **166** extending from the top surface into the gap or notch formed in the top surface. The tab **166** can assist in positioning and/or aligning the top backshell half relative to a bottom backshell half during assembly of the connector. Although FIG. 3

illustrates the top surface of the subassembly **150**, it should be understood that the bottom/opposing surface of the subassembly **150** can be a mirror image of the top surface of the subassembly **150**. In some embodiments, the depth of the opening **138** in the front housing **104** can be reduced to prevent latching/coupling of the plug housing **152** with the front housing **104** during use.

(25) The side walls **156**, **158**, bottom wall **160**, and top wall sections **162**, **164** define an internal space configured to at least partially receive the internal brace **154**. The proximal or end face of the plug housing **152** can include one or more extensions or flanges **168-174** protruding from the walls **156**, **158**, **160** and wall sections **162**, **164**. For example, the flanges **168**, **170** can extend from the proximal corners of the top wall sections **162**, **164**, and the flanges **172**, **174** can extend from the proximal edges of the side walls **156**, **158**.

(26) The internal brace **154** includes a planar proximal section **176** and includes a guiding block **178** at a proximal end. The guiding block **178** includes peaks **180** and valleys **182** configured to at least partially receive and/or support ferrules **184** and/or fibers from a fiber cable to be terminated. It should be understood that the internal brace **154** has the same guiding block **178** disposed on the opposing bottom surface of the internal brace **154**, with each guiding block **178** configured to support four fibers of a fiber cable (e.g., a mirror image of the top surface of the internal brace **154**). Although not illustrated, at this stage of the assembly process, the connector would be terminated to a fiber cable, and four fibers each would be routed to the individual fiber positions above and below the internal brace **154**, connecting with the respective ferrules **184**.

(27) The internal brace **154** can define a substantially planar proximal edge **186**, and the opposing side edges **188**, **190** can each include protrusions or flanges **192**, **194** extending perpendicularly therefrom. The flanges **192**, **194** can be spaced from each other, with the flanges **192** positioned at or near a midpoint of the section **176** and the flanges **194** positioned at or near the proximal edge **186**. During insertion of the internal brace **154** into the plug housing **152**, the flanges **192** can abut the proximal face of the plug housing **152**, thereby acting as a stop to ensure proper positioning of the internal brace **154** during assembly. As illustrated in FIG. 3, the subassembly **150** is inserted into the opening **138** of the adapter **100** before proceeding to the next step of assembly of the fiber connector.

(28) FIGS. 4-7 are perspective, top, side and rear views of a base station **200** of the present disclosure. The base station **200** includes features and/or contours configured to receive a backshell half and the adapter **100** loaded with the subassembly **150**. The base station **200** includes a distal or primary section **202** and a proximal or secondary section **204** extending from the section **202**. The section **202** includes a base **206** defining a substantially planar surface, side walls **208**, **210** extending perpendicularly on opposing sides of the base **206**, and a distal or rear wall **212** connecting the side walls **208**, **210**. The base **206** and walls **208-212** define a hollow interior configured to at least partially receive the loaded adapter **100**. The front surface or face **214** of the section **202** includes an opening extending into the hollow interior.

(29) The section **202** includes two inwardly facing protrusions **216**, **218** extending from the inner surface of the side walls **208**, **210**. The protrusions **216**, **218** are aligned with the front face **214** of the section **202**, and extend towards each other at the opening leading to the hollow interior. The section **202** includes a raised block **220**, **222** at each of the rear or distal corners extending from the base **206**. The blocks **220**, **222** are positioned at the intersection of the side wall **208** and the rear wall **212**, and the side wall **220** and the rear wall **212**, and define a narrower gap between the blocks **220**, **222** as compared to the space between the walls **208**, **210**. The protrusions **216**, **218** and blocks **220**, **222** can extend an equal partial height of the side walls **208**, **210**. The section **202** includes a raised step **224** formed in the base **206** at or near the face **214**, defining the connection to the section **204**.

(30) The planar surface area of the base **206**, the contours provided by the inner surfaces of the walls **208-212**, and the contours of the raised blocks **220**, **222** are configured and dimensioned complementary to the outer surfaces of the adapter **100**. (See, e.g., FIG. 9). For example, the

adapter **100** can be loaded into the section **202** such that the rear face **114** of the rear housing **102** abuts the rear wall **212**, and the front face **132** (or bottom edge of the front face **132**) of the front housing **104** abuts the raised step **224**, ensuring proper positioning of the adapter **100** within the base station **200**.

(31) The section **202** includes a lid or cover **226** hingedly coupled to the rear wall **212**. The rear wall **212** includes a cutout **228** configured to movably or rotatably receive a pivot section **230** extending from a body **232** of the cover **226**. The body **232** of the cover **226** generally defines a planar top surface **234** and a planar opposing bottom surface **236**. The front face or edge of the cover **226** includes downwardly directed flanges **238**, **240**. When the cover **226** is pivoted into the closed position (see, e.g., FIG. **10**), the flanges **238**, **240** substantially align with and are positioned over the protrusions **216**, **218**. The cover **226** and flanges **238**, **240** are dimensioned such that when the cover **226** is in the closed position, the inner surfaces of the flanges **238**, **240** abut against the front face **132** of the front housing **104**, thereby securing the proper position of the adapter **100** within the base station **200**. In the closed position, the bottom surface **236** abuts against or is positioned immediately over the top surface **242** of the section **202**. The cover **226** includes a U-shaped central cutout **244** at the front face and formed between the flanges **238**, **240**.

(32) The section **204** extends perpendicularly from the first section **202** and is aligned along a central longitudinal axis **246** of the base section **200**. The section **204** generally defines a planar, substantially rectangular body with a top surface **248**, an opposing bottom surface **250**, a front face **252**, and opposing side walls **254**, **256**. From the raised step **224** (e.g., a transition section), the top surface **248** generally defines an equally planar surface up to the edge of the front face **252**. On either side of the raised step **224** and adjacent to the extensions **216**, **218**, the section **204** includes raised platforms **258**, **260** elevated above the top surface **248** and forming an L-shaped raised step **262**. The platforms **258**, **260** are separated from either other by a gap **263** along the top surface **248** and extending up to the raised step **224**. The top surface **248** is therefore positioned higher than the base **206** surface, and the top surface of the platforms **258**, **260** is positioned higher than the top surface **248** of the section **204**.

(33) The section **204** includes a backshell support **264** disposed on the top surface **248** at or near the front face **252**. The backshell support **264** includes a cable support section **266** at a proximal end and two walls **268**, **270** extending outwardly and distally from the cable support section **266**. The space between the walls **268**, **270** is therefore dimensioned smaller at the connection to the cable support section **266**, and greater at the distal ends of the walls **268**, **270**. The backshell support **264** therefore defines a substantially Y-shaped configuration when viewed from above. The cable support section **266** includes a distal or rear face **272** and an opposing proximal or front face **274**. The inner surfaces of the walls **268**, **270** and the rear face **272** define a substantially triangular or trapezoidal space complementary to outer walls of a backshell half. The cable support section **266** includes a semi-circular groove **276** formed in the top surface and extending between the front and rear faces **274**, **272**. The groove **276** is configured to receive and support a fiber cable.

(34) FIG. **8** is a perspective view of the base station **200** loaded with a backshell half **280**. The backshell half **280** includes a base **282** with opposing side walls **284**, **286**, the base **282** and walls **284**, **286** defining a space configured and dimensioned to at least partially receive the terminated plug subassembly **150**. The walls **284**, **286** include upwardly directed extensions and/or latching flanges for engagement with opposing complementary extensions, latching flanges and/or grooves of a second backshell half. A distal edge **288** of the base includes a latching assembly **290** extending therefrom. The latching assembly **290** defines a smaller width than the width of the base **282**, and is substantially complementary to the gap **263** formed between the raised platforms **258**, **260** of the base station **200**. The backshell half **280** includes raised proximal walls **292**, **294** that define inwardly angled/narrowing outer surfaces and outwardly curved/convex inner surfaces.

(35) The outer surfaces of the walls **292**, **294** substantially correspond with the outwardly splayed configuration of the walls **268**, **270** of the backshell support **264**. The walls **268**, **270** narrow up to a

proximal face **296** of the backshell half **280**. The top surface of the walls **292**, **294** includes an inwardly formed groove **298** defining a crimping section for the fiber cable. During assembly, as illustrated in FIG. **8**, the backshell half **280** can be positioned onto the top surface **248** of the base station **200** such that the latching assembly **290** extends within the gap **263** and the proximal face **296** abuts the rear face **272** of the cable support **266**. The raised steps **262** of the platforms **258**, **260**, and the walls **268**, **270** of the backshell support **264**, ensure alignment of the backshell half **280** along the central longitudinal axis **246** of the base station **200**.

(36) FIG. **9** is a perspective view of the base station **200** loaded with the backshell half **280** and the adapter **100** assembled with the plug subassembly **150**. After loading of the backshell half **280** into the base station **200**, the adapter **100** and plug subassembly **150** can be loaded into the section **202** of the base station **200**. The rear wall **214** of the adapter **100** abuts against the inner surface of the rear wall **212** of the base station **200**, the side walls **110**, **112** abut against respective blocks **220**, **220**, and the front face **132** of the adapter **100** abuts against the raised step **224** to maintain alignment of the adapter **100** along the central longitudinal axis **246** of the base station **200**. The contours and components of the base station **200** therefore ensure accurate and repeatedly consistent alignment of the adapter **100** (and the plug subassembly **150**) relative to the backshell half **280**.

(37) The plug housing **152** of the subassembly **150** is positioned over the backshell half **280**, and the internal brace **154** is positioned between the walls **184**, **186** of the backshell half **280** (e.g., within the hollow interior of the backshell half **180**). The flanges **192**, **194** on opposing sides of the internal brace **154** can assist with alignment or positioning of the subassembly **150** relative to the backshell half **280**. The fibers associated with the fiber cable (not shown) terminated to the plug housing **152** can be routed through the interior passage of the backshell half **280**, through the crimping section defined by the groove **298** of the backshell half **280**, and along the groove **276** formed in the base section **200**. A crimp anchor **299** can be positioned partially within the groove **298** and partially within the groove **276**, and around the fiber cable, to allow for crimping of the fiber cable relative to the backshell half **280**.

(38) With the adapter **100**, the subassembly **150**, and the backshell half **280** properly positioned and loaded into the base section **200**, as illustrated in FIG. **10**, the cover **226** of the base section **200** is pivoted into a closed position to encase and fully cover the adapter **100**. As discussed previously, the inwardly facing surfaces of the extensions **238**, **240** abut the front face **132** of the adapter **100**, thereby preventing undesired movement of the adapter **100** (and the subassembly **150**) relative to the base station **200** and the backshell half **280**. In particular, by maintaining the fixed position of the adapter **100**, the subassembly **150** and the backshell half **280** are prevented from movement. The structure of the base station **200** and the adapter **100** (collectively referred to herein as the system) therefore ensure repeatedly consistent and accurate alignment of the assembly components.

(39) In some embodiments, a positioning tool **300** (e.g., a hand tool) of FIGS. **11-13** can be used to assist with ensuring that the internal brace **154** is properly located relative to the backshell half **280**. The tool **300** generally includes a body **302** defining the handle for maneuvering the tool **300**, and an elongated section **304** protruding from the body **302** to define the tip of the tool **300**. The body **302** includes an indicator **306** (e.g., text) on one surface to indicate to the operator the proper orientation of the tool **300**. The body **302** includes a proximal end **308** and a distal end **310**, with the section **304** extending from the distal end **310**.

(40) The tip of the section **304** includes a substantially V-shaped cutout **312** defining inwardly directed tapered surfaces **314**, **316**, and opposing tips **318**, **320**. The bottom tapered surface **316** can define a greater surface area than the surface area of the top tapered surface **314**. In some embodiments, the surface **316** can be dimensioned longer than the surface **314** as measured from the connection between the surfaces **314**, **316** and the respective tips **318**, **320**. The greater surface area and/or longer structure of the surface **316** can reduce potential slippage when the tool **300** is used to push against the rear surface or edge **186** of the internal brace **154**. In particular, the cutout

312 forms a space complementary to the edge **186** of the internal brace **154**.

(41) As illustrated in FIG. **13**, the cutout **312** can be positioned over the edge **186**, and the tool **300** can be carefully maneuvered to ensure proper positioning or seating of the internal brace **154** relative to the backshell half **280**. For example, the tool **300** allows for careful force to be applied to the edge **186** of the internal brace **154**, preventing potential damage to fibers passing over the internal brace **154**. Once the operator has positioned the internal brace **154** in the desired position, the tool **300** can be removed. It should be noted that use of the tool **300** is optional and is only needed if adjustment of the internal brace **154** position is desired.

(42) After inspection of the subassembly, the operator can snap a second backshell half relative to the backshell half **280** by aligning the corresponding latching flanges/grooves. For example, FIG. **14** shows a fully assembled fiber connector **400** including the plug subassembly **150**, the backshell half **280**, and a second backshell half **402** coupled to the backshell half **280**. The connector **400** includes a cable boot **404** that can be slid over a crimped fiber cable extending from the backshell halves **280**, **402**. Ferrules **406** extend from the proximal end of the connector **400**. The exemplary system discussed herein therefore provides for a convenient, simplified and accurate means for assembling internal components of the connector **400**, reducing the risk of misalignment/damage to internal components and/or erroneous assembly.

(43) While exemplary embodiments have been described herein, it is expressly noted that these embodiments should not be construed as limiting, but rather that additions and modifications to what is expressly described herein also are included within the scope of the invention. Moreover, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations, even if such combinations or permutations are not made express herein, without departing from the spirit and scope of the invention.

Claims

1. A system for fiber connector assembly, the system comprising: an adapter configured to at least partially receive a plug housing subassembly; and a base station configured to receive the adapter, the plug housing subassembly, and a first backshell component; wherein the base station includes structural features designed to align and maintain a position of the plug housing subassembly relative to the first backshell component prior to securing a second backshell component to the first backshell component; wherein the plug housing subassembly, the first backshell component and the second backshell component define an assembled fiber connector; and wherein the assembled fiber connector is configured to be removed from the adapter after the first backshell component and the second backshell component are coupled to the plug housing subassembly.
2. The system of claim 1, wherein the plug housing subassembly is a terminated plug housing subassembly including a plug housing and an internal brace.
3. The system of claim 1, wherein the adapter includes a rear housing engaged with a front housing, and wherein the front housing includes a body with an opening formed in a front face of the body and extending into a hollow interior of the body.
4. The system of claim 3, wherein the front housing includes an array of alignment sleeves extending from a rear wall within the hollow interior of the body, each of the alignment sleeves of the array of alignment sleeves configured to receive and support a fiber ferrule.
5. The system of claim 4, wherein each alignment sleeve of the array of alignment sleeves includes an inner surface fabricated from, covered by or coated with zirconia ceramic or phosphor bronze.
6. The system of claim 1, wherein the base station includes a primary section and a secondary section extending from the primary section, the primary section defining an enclosure configured to at least partially receive the adapter.
7. The system of claim 6, wherein the primary section includes a base with raised blocks extending

vertically on opposing sides of the base.

8. The system of claim 7, wherein the raised blocks define a narrowed section within the enclosure complementary to outer surfaces of the adapter.

9. The system of claim 6, wherein the base station includes a raised step along a base surface at a transition between the primary section and the secondary section.

10. The system of claim 9, wherein the raised step is configured to abut a front face of the adapter and an inner surface of a rear wall of the primary section is configured to abut a rear face of the adapter during assembly of the adapter with the base station.

11. The system of claim 6, wherein the secondary section includes a backshell support including a cable support section and outwardly flanged walls extending from the cable support section.

12. The system of claim 11, wherein the outwardly flanged walls define interior surfaces complementary to outer surfaces of the first backshell component.

13. The system of claim 11, wherein the outwardly flanged walls face the primary section of the base station.

14. The system of claim 6, wherein the secondary section includes raised platforms on opposing sides of a top surface, the raised platforms spaced by a gap.

15. The system of claim 14, wherein the gap is dimensioned complementary to a section of the first backshell component to ensure alignment of the first backshell component relative to the base station.

16. The system of claim 6, wherein the base station includes a cover pivotably coupled to the primary section, the cover configured to abut a front face of the adapter to maintain alignment and positioning of the adapter and the plug housing subassembly relative to the first backshell component.

17. The system of claim 16, wherein pivoting of the cover into a closed position over the adapter prevents movement of the adapter, the plug housing subassembly, and the first backshell component relative to the base station.

18. The system of claim 1, comprising a tool for manipulating a position of one or more components of the plug housing subassembly at least partially received by the adapter, the tool including a handle and a distal end with a V-shaped cutout complementary to an edge of the one or more components of the plug housing subassembly.

19. A system for fiber connector assembly, the system comprising: an adapter configured to at least partially receive a plug housing subassembly; a first backshell component configured to at least partially receive the plug housing subassembly; a second backshell component configured to at least partially receive the plug housing subassembly; and a base station configured to receive the adapter, the plug housing subassembly, and the first backshell component; wherein the base station includes structural features designed to align and maintain a position of the adapter and the plug housing subassembly relative to the first backshell component prior to securing the second backshell component to the first backshell component; wherein the plug housing subassembly, the first backshell component and the second backshell component define an assembled fiber connector; and wherein the assembled fiber connector is configured to be removed from the adapter after the first backshell component and the second backshell component are coupled to the plug housing subassembly.

20. A method of fiber connector assembly, comprising: at least partially positioning a plug housing subassembly into an adapter of a fiber connector assembly system; positioning a first backshell component of a fiber connector onto the base station of the fiber connector assembly system; positioning the adapter and the plug housing subassembly onto the base station; aligning a position of the plug housing subassembly relative to the first backshell component; and maintaining the aligned position of the plug housing subassembly relative to the first backshell component with structural features of the base station prior to securing a second backshell component to the first backshell component; wherein the plug housing subassembly, the first backshell component and the

second backshell component define an assembled fiber connector; and wherein the assembled fiber connector is configured to be removed from the adapter after the first backshell component and the second backshell component are coupled to the plug housing subassembly.
